

 c0c25013a5 / ProjExD_2





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 c0c25013a5 改行の数を修正#1 ecce131 · 26 minutes ago 

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```
1  import os
2  import sys
3  import pygame as pg
4  import random
5  import time
6
7  WIDTH, HEIGHT = 1100, 650
8  Δ={pg.K_UP:  (0, -5),
9      pg.K_DOWN: (0, +5),
10     pg.K_LEFT: (-5, 0),
11     pg.K_RIGHT: (+5, 0),
12     }
13  os.chdir(os.path.dirname(os.path.abspath(__file__)))
14
15  Δ_dict = {
16      (0,0): pg.transform.rotozoom(pg.image.load("fig/3.png"), 0, 1.0),
17      (-5,-5): pg.transform.rotozoom(pg.image.load("fig/3.png"), 45, 1.0),
18      (-5,0): pg.transform.rotozoom(pg.image.load("fig/3.png"), 0, 1.0),
19      (-5,+5): pg.transform.rotozoom(pg.image.load("fig/3.png"), -45, 1.0),
20      (0,+5): pg.transform.rotozoom(pg.transform.flip(pg.image.load("fig/3.png"), False, True), 90, 1.0),
21      (+5,+5): pg.transform.rotozoom(pg.image.load("fig/3.png"), -125, 1.0),
22      (+5,0): pg.transform.rotozoom(pg.transform.flip(pg.image.load("fig/3.png"), True, False), 0, 1.0),
23      (+5,-5): pg.transform.rotozoom(pg.image.load("fig/3.png"), 0, 1.0),
24      (0,-5): pg.transform.rotozoom(pg.image.load("fig/3.png"), 0, 1.0),
25  }
26
27  def check_bound(rct:pg.Rect) -> tuple[bool, bool]:
28      """
29      引数:効果トンまたは爆弾のRect
30      返り値:タプル（横方向判定結果、縦方向判定結果）
31      画面内ならTrue、画面外ならFalse
32      """
33      yoko, tate = True, True
34      if rct.left < 0 or rct.right > WIDTH:
35          yoko=False
36      if rct.top < 0 or rct.bottom > HEIGHT:
37          tate=False
38      return yoko, tate
39
40
41  def gameover(screen :pg.Surface) -> None:
```

```

42     """
43     ゲームオーバー画面を表示する関数
44     """
45     #背景
46     go_img = pg.Surface((WIDTH, HEIGHT))
47     pg.draw.rect(go_img, (0, 0, 0), (0, 0, WIDTH, HEIGHT))
48     go_img.set_alpha(100)
49     screen.blit(go_img, (0, 0))
50     #文字
51     go_font = pg.font.Font(None, 50)
52     txt = go_font.render("Game Over", True, (255, 255, 255))
53     screen.blit(txt, (WIDTH//2 - txt.get_width()//2, HEIGHT//2 - txt.get_height()//2))
54     #こうかとん
55     gokk_img = pg.image.load("fig/8.png")
56     screen.blit(gokk_img, (WIDTH//2 - txt.get_width()//2 - gokk_img.get_width(), HEIGHT//2 - gokk_img.get_height()//2))
57     screen.blit(gokk_img, (WIDTH//2 + txt.get_width()//2, HEIGHT//2 - gokk_img.get_height()//2))
58
59     pg.display.update()
60     time.sleep(5)
61
62
63     def init_bb_imgs() -> tuple[list[pg.Surface], list[int]]:
64         """
65         爆弾が時間とともに拡大、加速する関数
66         """
67         bb_imgs = []
68         bbaccs = [a for a in range(1,11)]
69         for r in range(1,11):
70
71             bb_img = pg.Surface((20*r, 20*r))
72             bb_img.set_colorkey((0, 0, 0))
73             pg.draw.circle(bb_img, (255, 0, 0), (10*r, 10*r), 10*r)
74             bb_imgs.append(bb_img)
75
76         return bb_imgs, bbaccs
77
78
79     def main():
80         pg.display.set_caption("逃げろ！こうかとん")
81         screen = pg.display.set_mode((WIDTH, HEIGHT))
82         bg_img = pg.image.load("fig/pg_bg.jpg")
83         kk_img = pg.transform.rotozoom(pg.image.load("fig/3.png"), 0, 0.9)
84         kk_rct = kk_img.get_rect()
85         kk_rct.center = 300, 200
86         bb_img = pg.Surface((20, 20))
87         pg.draw.circle(bb_img, (255, 0, 0), (10, 10), 10)
88         bb_img.set_colorkey((0, 0, 0))
89
90         bb_rct = bb_img.get_rect()
91         bb_rct.center = random.randint(0, WIDTH), random.randint(0, HEIGHT)
92         clock = pg.time.Clock()
93         tmr = 0
94
95         vx,vy= +5, +5
96         bb_imgs, bb_accs = init_bb_imgs()
97
98         while True:
99             for event in pg.event.get():

```

```
100         if event.type == pg.QUIT:
101             return
102
103         screen.blit(bg_img, [0, 0])
104         avx = vx*bb_accs[min(tmr//500,9)]
105         avy = vy*bb_accs[min(tmr//500,9)]
106         bb_rct.move_ip(avx, avy)
107         bb_img = bb_imgs[min(tmr//500,9)]
108         bb_rct.width = bb_img.get_rect().width
109         bb_rct.height = bb_img.get_rect().height
110         if check_bound(bb_rct) == (False, True):
111             vx *= -1
112         elif check_bound(bb_rct) == (True, False):
113             vy *= -1
114         screen.blit(bb_img, bb_rct)
115         #ゲームオーバー画面
116         if kk_rct.colliderect(bb_rct):
117             gameover(screen)
118
119         key_lst = pg.key.get_pressed()
120         sum_mv = [0, 0]
121         #if key_lst[pg.K_UP]:
122         #    sum_mv[1] -= 5
123         #if key_lst[pg.K_DOWN]:
124         #    sum_mv[1] += 5
125         #if key_lst[pg.K_LEFT]:
126         #    sum_mv[0] -= 5
127         #if key_lst[pg.K_RIGHT]:
128         #    sum_mv[0] += 5
129         for x,y in DELTA.items():
130             if key_lst[x]:
131                 sum_mv[0] += y[0]
132                 sum_mv[1] += y[1]
133
134
135         kk_rct.move_ip(sum_mv)
136         if check_bound(kk_rct) != (True, True):
137             kk_rct.move_ip(-sum_mv[0], -sum_mv[1])
138         screen.blit(kk_img, kk_rct)
139
140         pg.display.update()
141         tmr += 1
142         clock.tick(50)
143
144
145 if __name__ == "__main__":
146     pg.init()
147     main()
148     pg.quit()
149     sys.exit()
```